

# Trigonometry

## Week 1B

# Overview

## *Trigonometry*

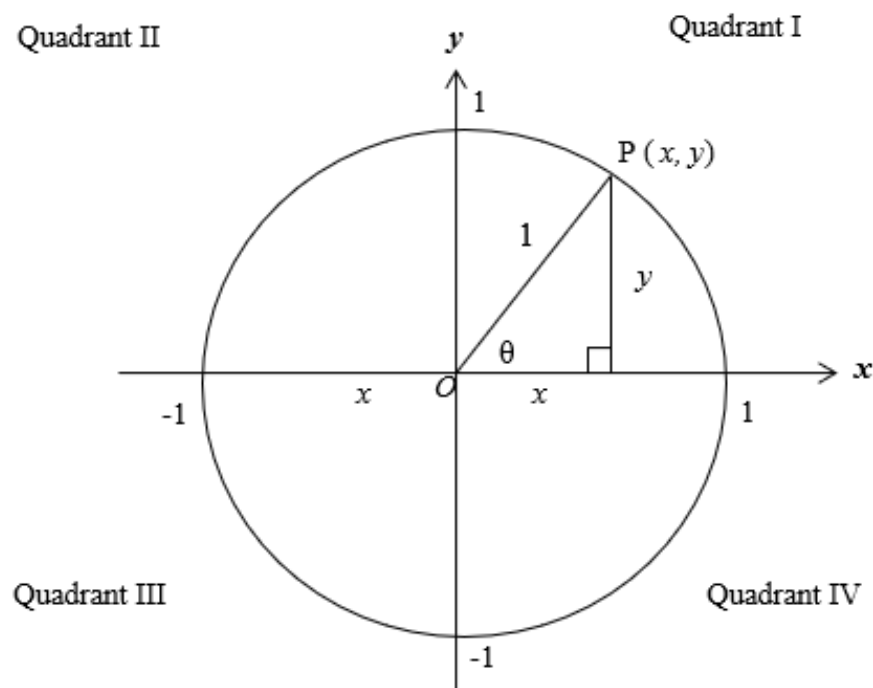
*Key Ideas - Angles of any Magnitude*

- Signs of Trigonometric Ratios*
- Calculating Trigonometric Ratios for Angles of any Magnitude in degrees*
- Solving trigonometric equations in degrees*

## Key Idea – Angles of any Magnitude

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If we are only using acute angles, then the definitions of the trigonometric ratios that we have studied so far would be sufficient. However, sometimes we use angles that are larger than acute angles or angles that are written as negative angles. Consider a unit circle (i.e. a circle with radius 1 unit) with centre  $O$  in the  $x$ - $y$  coordinate plane.  $P$  is a point on the unit circle. Let  $OP$  be a radius of that circle (i.e. a radius of 1 unit). Point  $P$  can be in any quadrant.



From the right-angled triangle

- $\sin \theta = \frac{y}{1} = y$
- $\cos \theta = \frac{x}{1} = x$

Therefore

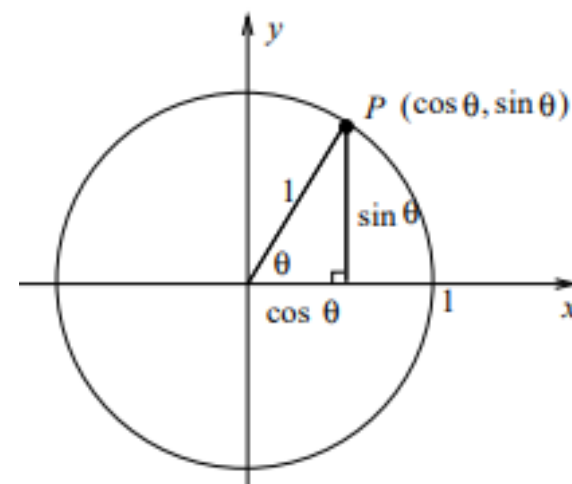
$$P(x, y) \rightarrow P(\cos x, \sin x)$$

This gives us a more general definition of the trigonometric ratios in a form suitable for angles of **any magnitude**.

$$\sin \theta = y \text{ coordinate of } P$$

$$\cos \theta = x \text{ coordinate of } P$$

$$\tan \theta = \frac{y}{x} = \frac{\sin \theta}{\cos \theta}$$

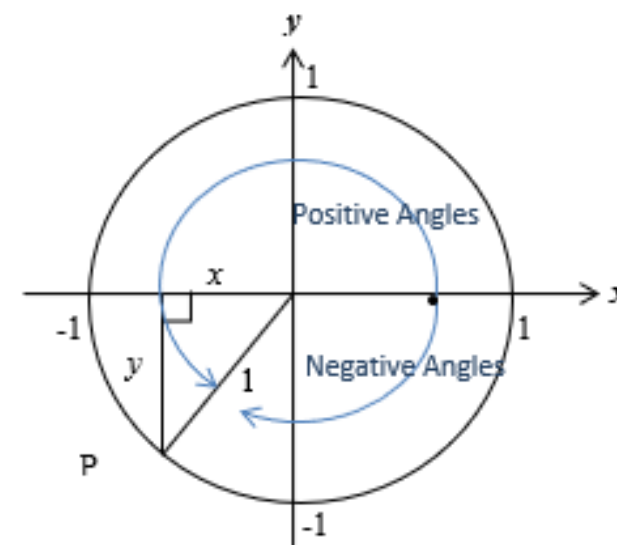


### Positive angles:

The angle  $\theta$  measured from the **positive**  $x$ -axis in an **anti-clockwise** direction to  $OP$  is a **positive** angle.

### Negative angles:

The angle  $\theta$  measured from the **positive**  $x$ -axis in a **clockwise** direction to  $OP$  is a **negative** angle



## Trigonometric ratios of $0^\circ$ , $90^\circ$ , $180^\circ$ , $270^\circ$ , $360^\circ$

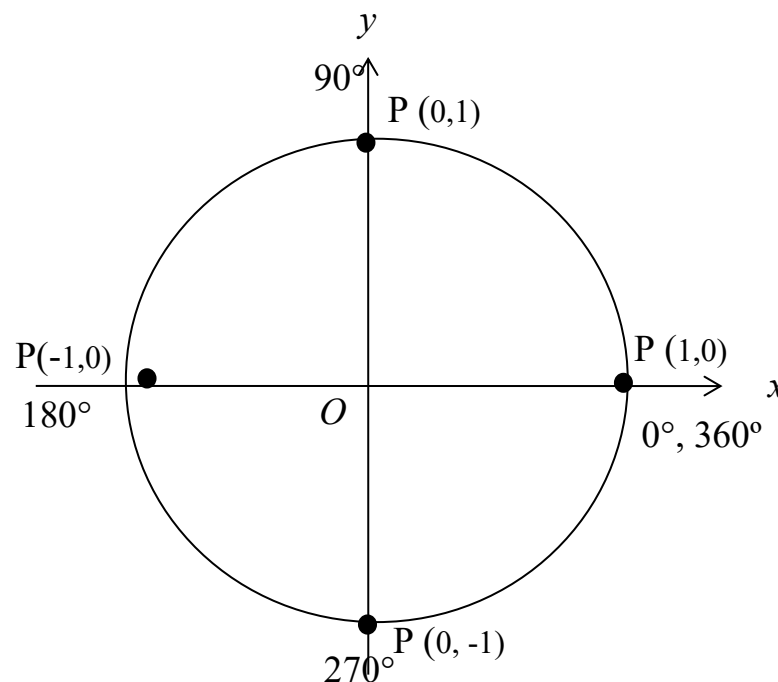
This wider definition allows us to calculate trigonometry ratios for angles that are 90 degrees or greater. The trigonometry ratios for these angles could not have been calculated previously as the right-angled triangle definition limits the use of the trigonometric ratios to angles between 0 and 90 degrees.

### Discussion Question:

Using the coordinates of  $P$  at various positions on the unit circle, complete the table below.

|               | $0^\circ$         | $90^\circ$                 | $180^\circ$        | $270^\circ$                 | $360^\circ$       |
|---------------|-------------------|----------------------------|--------------------|-----------------------------|-------------------|
| $\sin \theta$ | 0                 | 1                          | 0                  | -1                          | 0                 |
| $\cos \theta$ | 1                 | 0                          | -1                 | 0                           | 1                 |
| $\tan \theta$ | $\frac{0}{1} = 0$ | $\frac{1}{0} = \text{DNE}$ | $\frac{0}{-1} = 0$ | $\frac{-1}{0} = \text{DNE}$ | $\frac{0}{1} = 0$ |

DNE = does not exist



This information can be used to calculate the value of reciprocal trigonometric ratios as well as trigonometry ratios for negative angles.

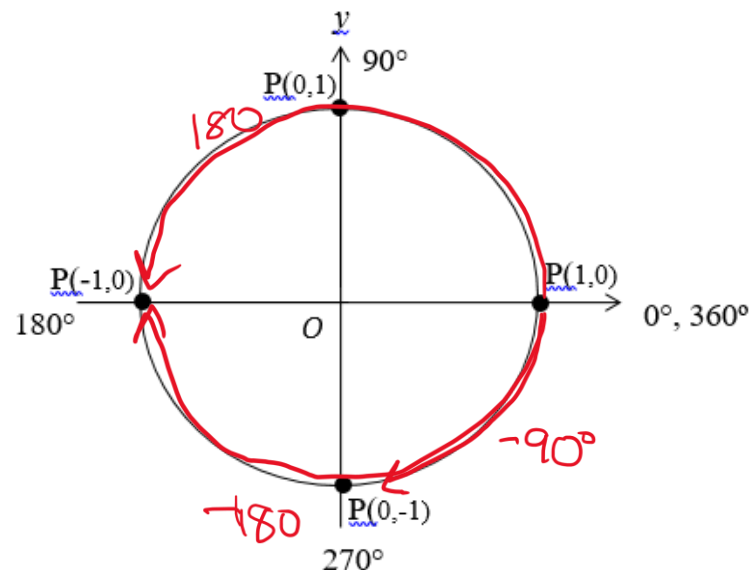
### Exercise:

Evaluate:

(a)  $\sin(-90^\circ) = -1$

(b)  $\cos(-180^\circ) = -1$

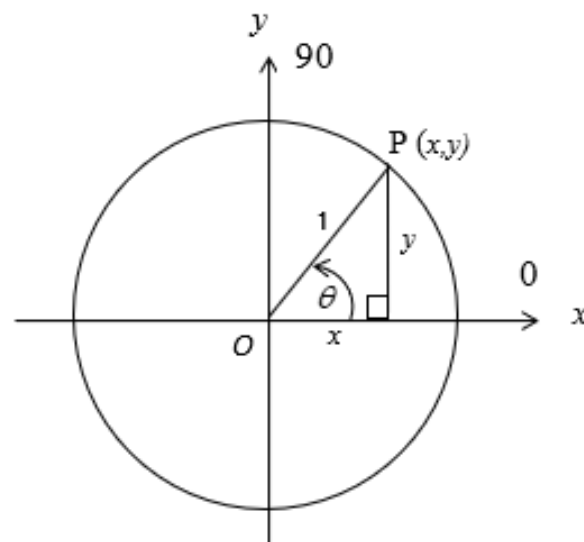
(c)  $\cot(180^\circ) = \frac{1}{\tan 180^\circ}$   
 $= \frac{1}{0}$   
 $= \frac{1}{0} = \text{DNE}$



## Key Idea – Signs of the Trigonometric Ratios

As the values of the  $\sin \theta$ ,  $\cos \theta$ ,  $\tan \theta$  ratios are related to the coordinates of point  $P$ , for angles greater than  $90^\circ$  it is possible to obtain trigonometric ratio values that are either positive or negative.

In **Quadrant 1**, we have acute angles: i.e. angles between  $0^\circ$  and  $90^\circ$ .



### Discussion Questions:

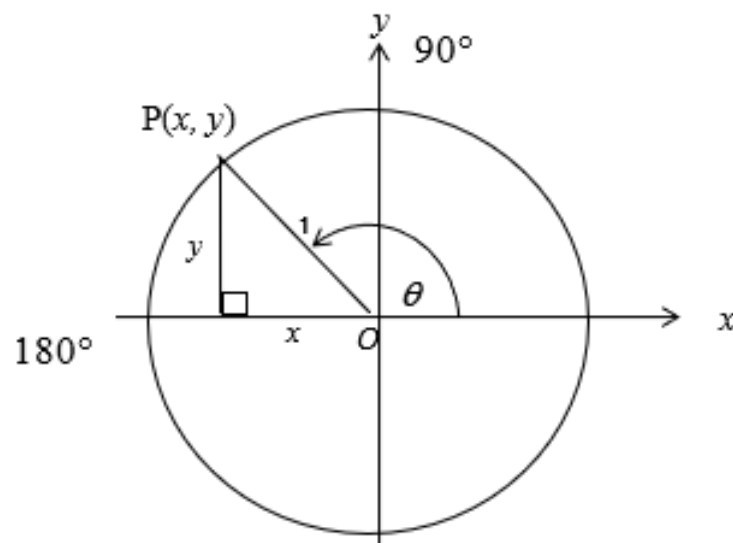
- (a) Would the  $x$  coordinate of point  $P$  be a positive or negative value?  $(+)$
- (b) Would the  $y$  coordinate of point  $P$  be a positive or negative value?  $(+)$
- (c) Use this information to state whether the  $\sin$ ,  $\cos$  and  $\tan$  ratios will be positive or negative.

$$\cos \theta = x \quad (+)$$

$$\sin \theta = y \quad (+)$$

$$\tan \theta = \frac{y}{x} \quad (+)$$

In **Quadrant 2**, we have angles between  $90^\circ$  and  $180^\circ$ .



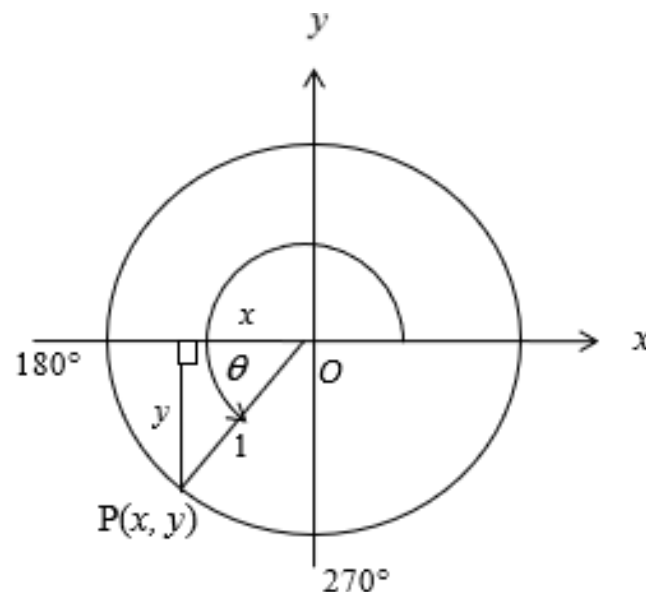
### Discussion Questions:

- (a) Would the  $x$  coordinate of point  $P$  be a positive or negative value?  $\ominus$
- (b) Would the  $y$  coordinate of point  $P$  be a positive or negative value?  $\oplus$
- (c) Use this information to state whether the sin, cos and tan ratios will be positive or negative.

$$\cos \theta = x \quad \ominus \quad \sin \theta = y \quad \oplus \quad \tan \theta = \frac{y}{x} = \frac{+}{-} \quad \ominus$$



In **Quadrant 3**, we have angles between  $180^\circ$  and  $270^\circ$ .

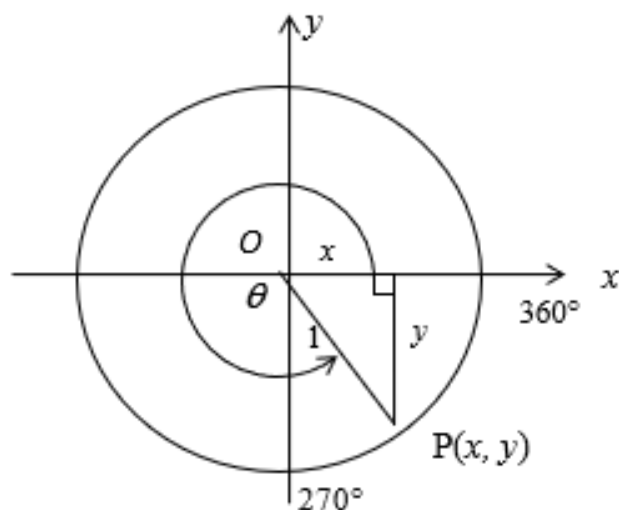


### Discussion Questions:

- (a) Would the  $x$  coordinate of point  $P$  be a positive or negative value?  $(-)$
- (b) Would the  $y$  coordinate of point  $P$  be a positive or negative value?  $(-)$
- (c) Use this information to state whether the sin, cos and tan ratios will be positive or negative.

$$\cos \theta = x \quad (-) \quad \sin \theta = y \quad (-) \quad \tan \theta = \frac{y}{x} = \frac{-}{-} \quad (+)$$

In **Quadrant 4**, we have angles between  $270^\circ$  and  $360^\circ$ .



**Discussion Questions:**

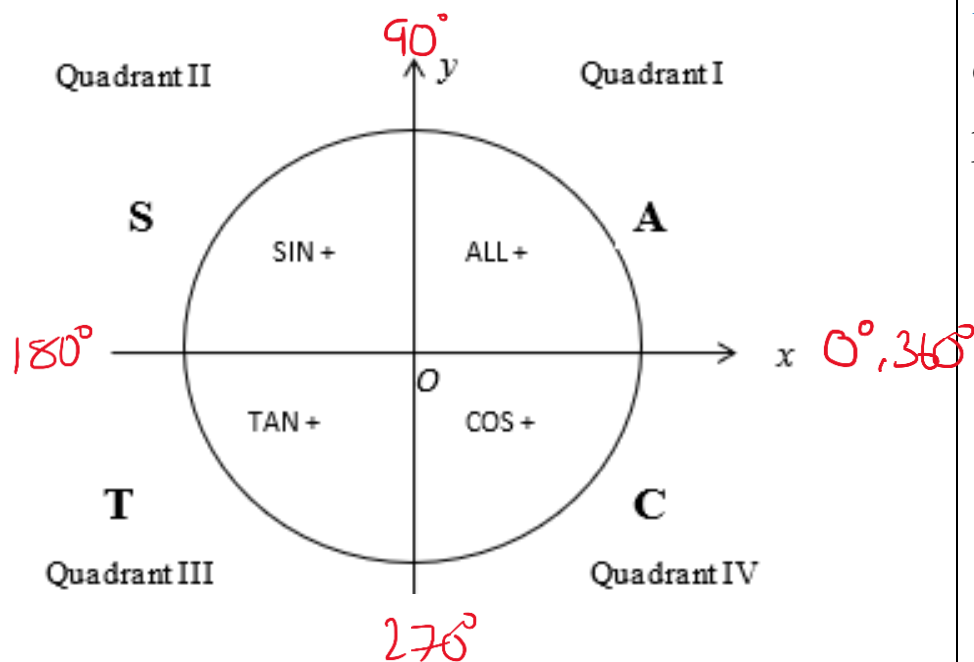
- (a) Would the  $x$  coordinate of point  $P$  be a positive or negative value?  $(+)$
- (b) Would the  $y$  coordinate of point  $P$  be a positive or negative value?  $(-)$
- (c) Use this information to state whether the sin, cos and tan ratios will be positive or negative.

$$\cos \theta = x \quad (+)$$

$$\sin \theta = y \quad (-)$$

$$\tan \theta = \frac{y}{x} = \frac{-}{+} \quad (-)$$

The following diagram summarises the positive ratios in each quadrant:



**Note:** This is often remembered using the mnemonic “All Stations To Central”.

**Exercise:** State the quadrant of each angle and determine whether the trigonometric ratio is positive or negative:

|                    | Quadrant | Sign |
|--------------------|----------|------|
| $\sin 160^\circ$   | 2        | +    |
| $\cos (-60^\circ)$ | 4        | +    |
| $\cos 240^\circ$   | 3        | -    |
| $\tan (-30^\circ)$ | 4        | -    |
| $\cos 420^\circ$   | 1        | +    |
| $\sin 280^\circ$   | 4        | -    |

**Note:** Acute Angles:  $0^\circ < \theta < 90^\circ$

Obtuse Angles:  $90^\circ < \theta < 180^\circ$

Reflex Angles:  $180^\circ < \theta < 360^\circ$

**Exercise:**

Complete the following table by determining the sign of the trigonometric ratios.

| Angle $\theta$ | $\sin \theta$ | $\cos \theta$ | $\tan \theta$ |
|----------------|---------------|---------------|---------------|
| $150^\circ$    | +             | -             | -             |
| $190^\circ$    | -             | -             | +             |
| $310^\circ$    | -             | +             | -             |
| $-100^\circ$   | -             | -             | +             |
| $400^\circ$    | +             | +             | +             |
| $800^\circ$    | +             | +             | +             |

## Key Idea – Calculating Trigonometric Ratios for Angles of any Magnitude (Degrees)

The concept of the unit circle allows us to calculate trigonometric ratio values for angles of any magnitude.

$$\sin \theta = y \text{ coordinate of } P \quad \cos \theta = x \text{ coordinate of } P \quad \tan \theta = \frac{y \text{ coordinate of } P}{x \text{ coordinate of } P}$$

### Investigate:

Using ratios:  $\cos 120^\circ = -0.5$  and  $\sin 120^\circ = 0.87$

the angle of  $120^\circ$  has been drawn on the unit circle.  $\rightarrow$

(a) Plot angle  $60^\circ$  on the unit circle.

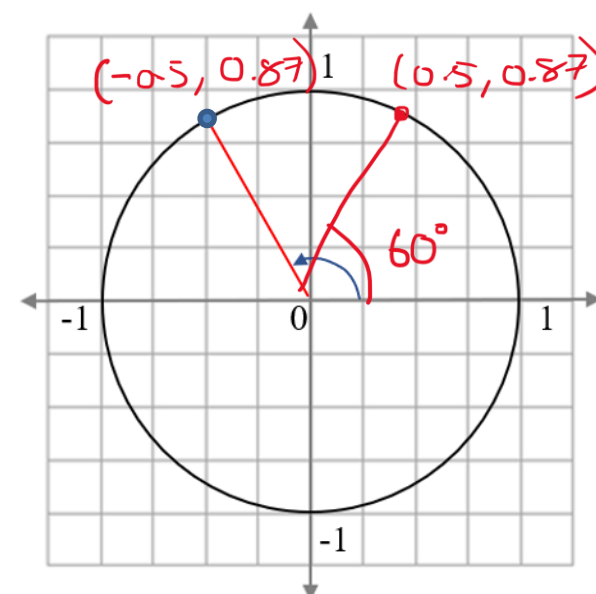
(b) Compare the values of:

$\cos 120^\circ$  and  $\cos 60^\circ$

$\sin 120^\circ$  and  $\sin 60^\circ$

$\tan 120^\circ$  and  $\tan 60^\circ$

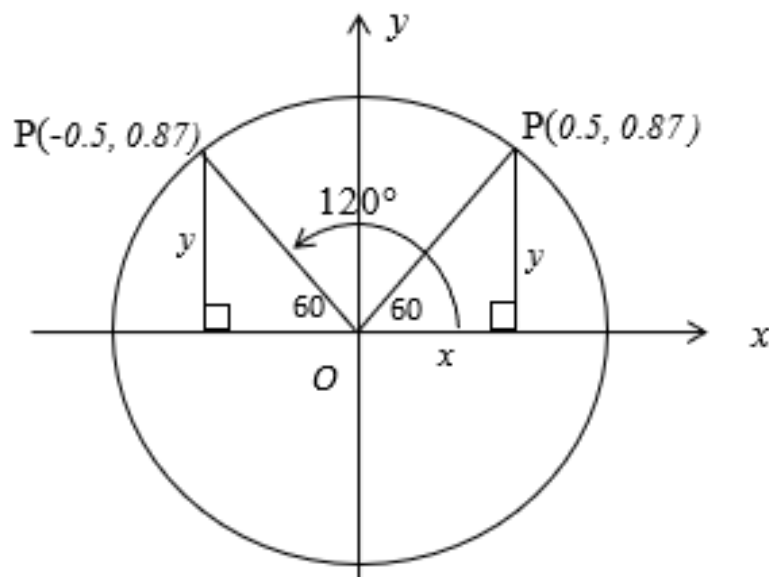
$$\begin{aligned} \cos 120^\circ &= -0.5 \\ \cos 60^\circ &= 0.5 \end{aligned} \rightarrow \text{opposite signs}$$
$$\begin{aligned} \sin 120^\circ &= 0.87 \\ \sin 60^\circ &= 0.87 \end{aligned} \rightarrow \text{same}$$
$$\tan 120^\circ = \frac{-0.5}{0.87}$$
$$\tan 60^\circ = \frac{0.5}{0.87} \rightarrow \text{opposite signs}$$



## Reference Angles (Degrees)

A “**reference angle**” is the acute angle (less than  $90^\circ$ ) formed between an angle and the  $x$ -axis. It helps us find trigonometric values for angles outside the first quadrant by linking them to an equivalent angle in Quadrant 1. The reference angle has the same trig value as the original angle, but the positive or negative sign depending on which quadrant the original angle is in.

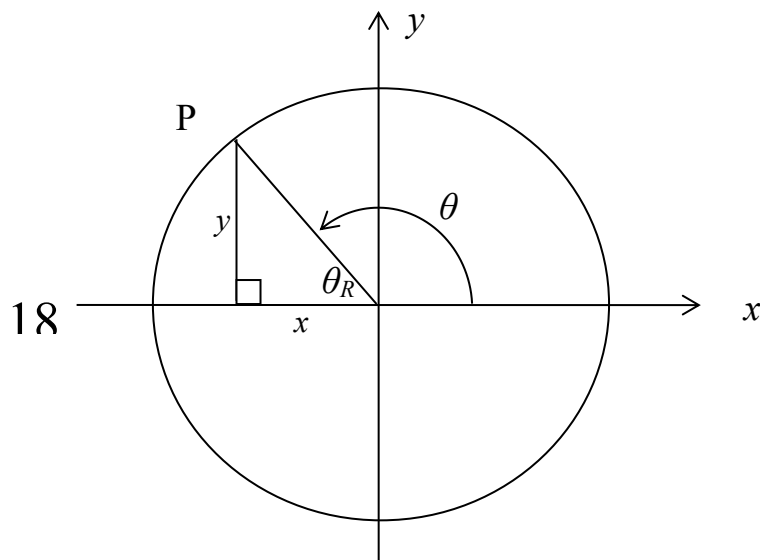
For example, in the previous activity we observed that  $\sin 120^\circ = \sin 60^\circ$  and  $\cos 120^\circ = -\cos 60^\circ$ . The trigonometric ratios of  $120^\circ$  are related to the trigonometric ratios of  $60^\circ$ . Therefore, the angle  $60^\circ$  is the reference angle for  $120^\circ$ .



In the first quadrant both the  $x$  and  $y$  coordinates are positive therefore all the trigonometric ratios of acute angles are positive.

## Second Quadrant Angles

Any angle in the second quadrant can be related to a first quadrant reference angle  $\theta_R^\circ$  using the following: Reference angle in the second quadrant:  $\theta_R = 180^\circ - \theta$



**Examples:** For the following obtuse angles, determine their reference angle:

(a)  $\theta = 120^\circ$

$$\begin{aligned}\theta_R &= 180^\circ - 120^\circ \\ &= 60^\circ\end{aligned}$$

$\theta_R = 60^\circ$  is the **reference** angle to  $120^\circ$

(b)  $\theta = 140^\circ$

$$\begin{aligned}\theta_R &= 180^\circ - 140^\circ \\ &= 40^\circ\end{aligned}$$

$\theta_R = 40^\circ$  is the **reference** angle to  $140^\circ$

## Third Quadrant Angles

Any angle in the third quadrant can be related to a first quadrant reference angle  $\theta_R^\circ$  using the following: Reference angle in the second quadrant:  $\theta_R = \theta - 180^\circ$

**Examples:** For the following reflex angles, determine their reference angle:

(a)  $\theta = 210^\circ$

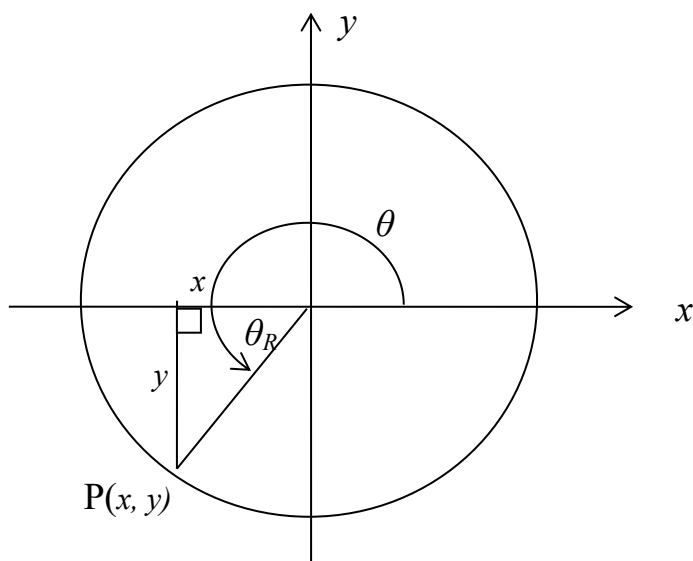
$$\begin{aligned}\theta_R &= 210^\circ - 180^\circ \\ &= 30^\circ\end{aligned}$$

$\theta_R = 30^\circ$  is the **reference** angle to  $210^\circ$

(b)  $\theta = 225^\circ$

$$\begin{aligned}\theta_R &= 225^\circ - 180^\circ \\ &= 45^\circ\end{aligned}$$

$\theta_R = 45^\circ$  is the **reference** angle to  $225^\circ$





## Fourth Quadrant Angles

Any angle in the fourth quadrant can be related to a first quadrant reference angle  $\theta_R^\circ$  using the following: Reference angle in the second quadrant:  $\theta_R = 360^\circ - \theta$

**Examples:** For the following reflex angles, determine their reference angle:

(a)  $\theta = 300^\circ$

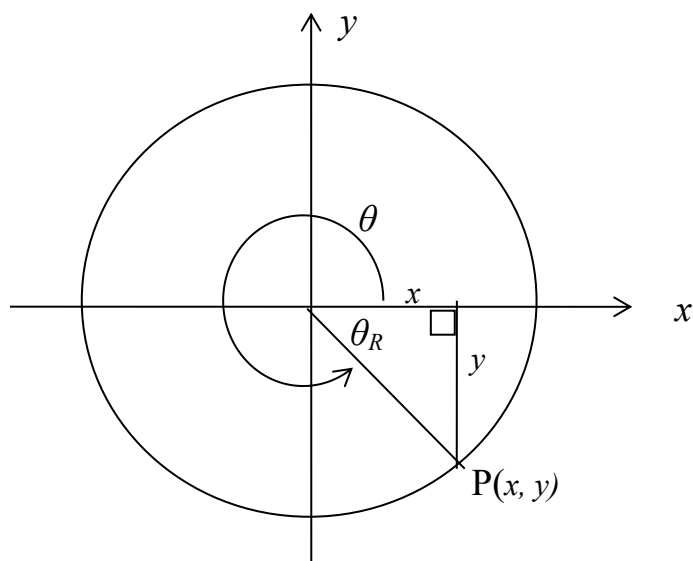
$$\begin{aligned}\theta_R &= 360^\circ - 300^\circ \\ &= 60^\circ\end{aligned}$$

$\theta_R = 60^\circ$  is the **reference** angle to  $300^\circ$

(b)  $\theta = 330^\circ$

$$\begin{aligned}\theta_R &= 360^\circ - 330^\circ \\ &= 30^\circ\end{aligned}$$

$\theta_R = 30^\circ$  is the **reference** angle to  $330^\circ$



## Finding reference angles

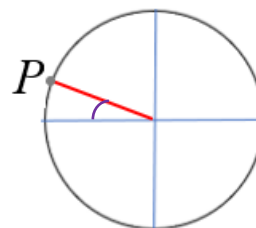
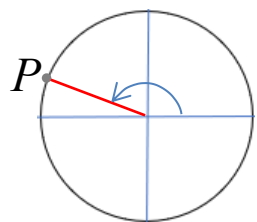
Step 1. Draw a circle showing the angle

Step 2. Find the reference angle between OP and the **x-axis**.

The reference angle is always positive

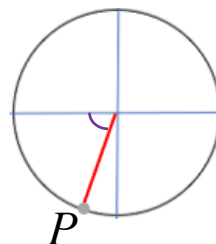
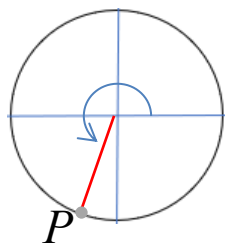
**Examples:** Find the reference angle for the following angles:

$160^\circ$



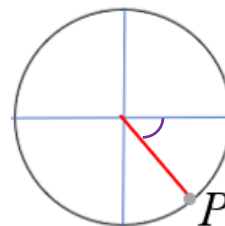
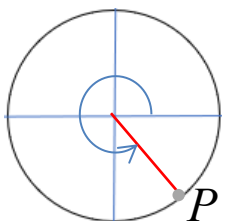
$$\begin{aligned}\theta_R &= 180 - 160 \\ &= 20^\circ\end{aligned}$$

$250^\circ$



$$\begin{aligned}\theta_R &= 250 - 180 \\ &= 70^\circ\end{aligned}$$

$320^\circ$

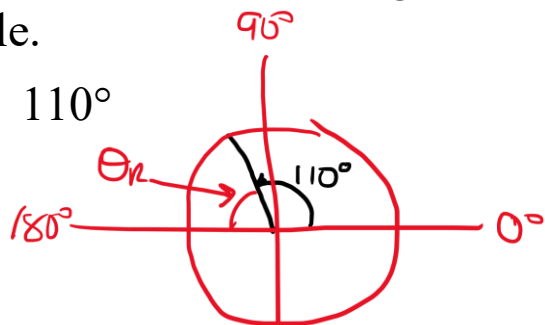


$$\begin{aligned}\theta_R &= 360 - 320 \\ &= 40^\circ\end{aligned}$$

### Exercise:

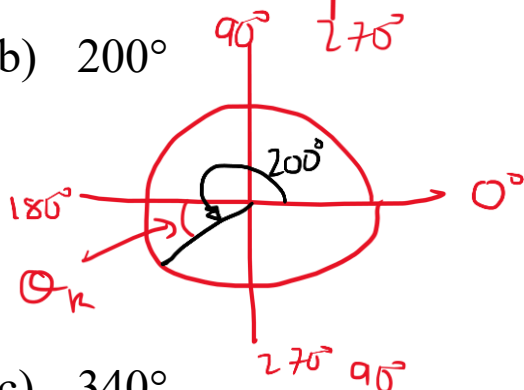
For each of the following, draw a circle showing the specified angle. Then find the reference angle.

(a)  $110^\circ$



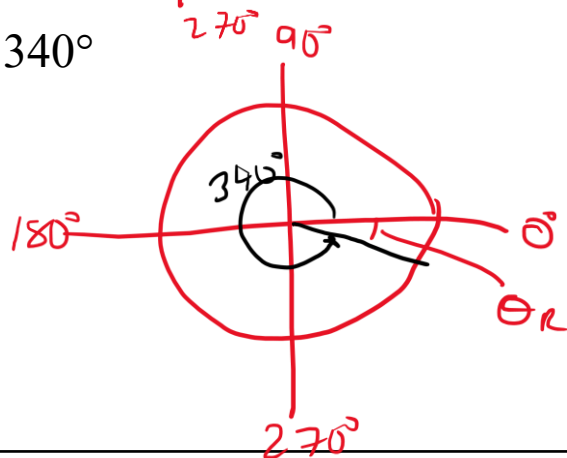
$$\begin{aligned}\theta_r &= 180^\circ - 110^\circ \\ &= 70^\circ\end{aligned}$$

(b)  $200^\circ$



$$\begin{aligned}\theta_r &= 200^\circ - 180^\circ \\ &= 20^\circ\end{aligned}$$

(c)  $340^\circ$



$$\begin{aligned}\theta_r &= 360^\circ - 340^\circ \\ &= 20^\circ\end{aligned}$$

## Exact Trigonometric Ratios for Angles of any Magnitude (Degrees)

Last week the exact trigonometric ratios of  $30^\circ$ ,  $45^\circ$  and  $60^\circ$  were found by using right-angled triangles.

| degrees | $30^\circ$           | $45^\circ$           | $60^\circ$           |
|---------|----------------------|----------------------|----------------------|
| sin     | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ |
| cos     | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        |
| tan     | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           |

When an angle in  $Q_2$ ,  $Q_3$  or  $Q_4$  has a reference angle of  $30^\circ$ ,  $40^\circ$  or  $60^\circ$ , the exact trigonometric ratios may be determined.

The steps involved for finding the exact trigonometric ratio are:

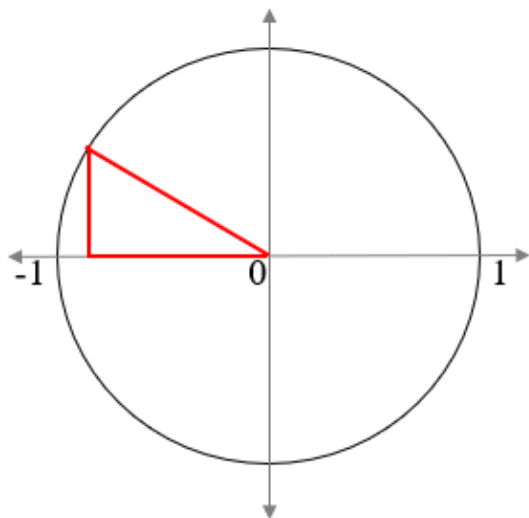
- Identify the quadrant in which the angle lies.
- Identify whether the trigonometric ratio is positive or negative in this quadrant.
- Determine the acute reference angle.
- Write the trigonometric ratio in terms of the reference acute angle.
- State the exact trigonometric ratio value.

**Example:** Find the exact value of:

$\tan 150^\circ$

- $150^\circ$  lies in the 2<sup>nd</sup> quadrant
- $\tan$  is **negative** in the 2<sup>nd</sup> quadrant
- $\theta_R = 180 - 150 = 30$

The reference angle is  $30^\circ$

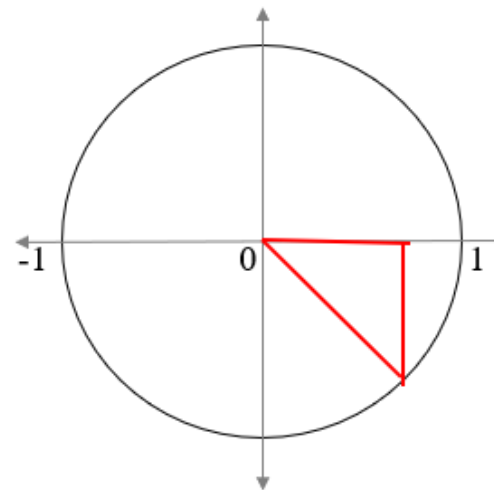


$$\begin{aligned}\tan 150^\circ &= -\tan 30^\circ \\ &= -\frac{1}{\sqrt{3}}\end{aligned}$$

$\cos 315^\circ$

- $315^\circ$  lies in the 4th quadrant
- $\cos$  is **positive** in the 4th quadrant
- $\theta_R = 360 - 315 = 45$

The reference angle is  $45^\circ$



$$\begin{aligned}\cos 315^\circ &= \cos 45^\circ \\ &= \frac{1}{\sqrt{2}}\end{aligned}$$

### Exercise:

Find the exact value of

(a)  $\cos 330^\circ$

- $330^\circ$  is in the 4<sup>th</sup> quadrant
- $\cos$  is positive in the 4<sup>th</sup> quadrant
- $\Theta_n = 360^\circ - 330^\circ = 30^\circ$

$$\begin{aligned}\cos 330^\circ &= \cos 30^\circ \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

(b)  $\sin 135^\circ$

- $135^\circ$  is in the 2<sup>nd</sup> quadrant
- $\sin$  is positive in the 2<sup>nd</sup> quadrant
- $\Theta_n = 180^\circ - 135^\circ = 45^\circ$

$$\begin{aligned}\sin 135^\circ &= \sin 45^\circ \\ &= \frac{1}{\sqrt{2}}\end{aligned}$$

(c)  $\tan(-60^\circ) = \tan(300^\circ)$

- $300^\circ$  is in the 4<sup>th</sup> quadrant
- $\tan$  is negative in the 4<sup>th</sup> quadrant
- $\Theta_n = 360^\circ - 300^\circ = 60^\circ$

$$\begin{aligned}\tan(-60^\circ) &= -\tan 60^\circ \\ &= -\sqrt{3}\end{aligned}$$

## Key Idea – Solving trigonometric equations in degrees

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Within the ranged specified, all angles that satisfy a trigonometric equation need to be identified.

To solve trigonometric equations:

Step 1: Determine the quadrants.

Step 2: Determine the reference (acute) angle.

Step 3: Determine the angles in the appropriate quadrants.

**Example:** Solve  $\cos \theta = -\frac{1}{2}$ ,  $0^\circ \leq \theta^\circ \leq 360^\circ$ .

Step 1     $\cos$  is negative in  $Q_2$  and  $Q_3$

Step 2    Reference angle is  $\theta_R = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$

Step 3    In  $Q_2$      $\theta = 180^\circ - 60^\circ = 120^\circ$

            In  $Q_3$      $\theta = 180^\circ + 60^\circ = 240^\circ$

### Example:

Angle  $\theta$  lies between  $0^\circ$  and  $360^\circ$ . Find all possible values for  $\theta$ , correct to the nearest minute, given  $\sin \theta = 0.6784$ .

Step 1  $\sin$  is positive in  $Q_1$  and  $Q_2$

Step 2 Reference angle is  $\theta_R = \sin^{-1} 0.6784 = 42^\circ 43'$

Step 3 In  $Q_1$   $\theta = 42^\circ 43'$

In  $Q_2$   $\theta = 180^\circ - 42^\circ 43' = 137^\circ 17'$

Therefore  $\theta = 42^\circ 43', 137^\circ 17'$

### Exercise:

Angle  $\theta$  lies between  $0^\circ$  and  $360^\circ$ . Find all possible values for  $\theta$ , correct to the nearest degree, given  $\tan \theta = -1.327$

- $\tan$  is negative in  $Q_2$  &  $Q_4$
- reference angle  $= \theta_R = \tan^{-1}(1.327) = 53^\circ$
- in  $Q_2 \rightarrow \theta = 180^\circ - 53^\circ = 127^\circ$
- in  $Q_4 \rightarrow \theta = 360^\circ - 53^\circ = 307^\circ$



$\therefore \theta = 127^\circ, 307^\circ$



### Exercise:

Solve each equation for  $0^\circ \leq \theta \leq 360^\circ$ . Where required, round answer to 2 decimal places.

(a)  $\cos \theta = \frac{\sqrt{3}}{2}$

•  $\cos$  is positive in Q1 & Q4

•  $\theta_n = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 30^\circ$

• in Q1  $\rightarrow \theta = 30^\circ$

in Q4  $\rightarrow \theta = 360^\circ - 30^\circ$   
 $= 330^\circ$

(b)  $\sin \theta = -\frac{1}{\sqrt{2}}$

•  $\sin$  is negative in Q3 & Q4

•  $\theta_n = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$

• in Q3  $\rightarrow \theta = 180^\circ + 45^\circ = 225^\circ$   
in Q4  $\rightarrow \theta = 360^\circ - 45^\circ = 315^\circ$

(c)  $1 + 2\sin \theta = 6\sin \theta$

$6\sin \theta - 2\sin \theta = 1$

$4\sin \theta = 1$

$\sin \theta = \frac{1}{4}$

•  $\sin$  is positive in Q1 & Q2

•  $\theta_n = \sin^{-1}\left(\frac{1}{4}\right) = 14.48^\circ$

• in Q1  $\rightarrow \theta = 14.48^\circ$

in Q2  $\rightarrow \theta = 180^\circ - 14.48^\circ$   
 $= 165.52^\circ$

(d)  $\cot \theta + \sqrt{3} = 0$

$\cot \theta = -\sqrt{3}$

$\tan \theta = -\frac{1}{\sqrt{3}}$

•  $\tan$  is negative in Q2 & Q4

•  $\theta_n = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ$

• in Q2  $\rightarrow \theta = 180^\circ - 30^\circ = 150^\circ$

in Q4  $\rightarrow \theta = 360^\circ - 30^\circ = 330^\circ$

## Further Trigonometric equations

**Example:** Solve each equation for  $0^\circ \leq \theta \leq 360^\circ$ .

$$\sec^2 \theta = 2$$

$$\sec \theta = \pm \sqrt{2}$$

$$\cos \theta = \pm \frac{1}{\sqrt{2}}$$

This gives values of  $\theta$  in all 4 quadrants

$$\text{In the first quadrant } \cos \theta = \frac{1}{\sqrt{2}} \rightarrow \theta_R = 45^\circ$$

$$\text{Therefore } \theta = 45^\circ, 180^\circ \pm 45^\circ, 360^\circ - 45^\circ$$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

$$\sin^2 \theta - \sin \theta = 2$$

$$\sin^2 \theta - \sin \theta - 2 = 0$$

$$(\sin \theta - 2)(\sin \theta + 1) = 0$$

$$\text{Therefore } (\sin \theta - 2) = 0 \text{ or } (\sin \theta + 1) = 0$$

$$\sin \theta - 2 = 0 \quad \text{or} \quad \sin \theta + 1 = 0$$

$$\sin \theta = 2 \quad \sin \theta = -1$$

$$\text{No solution} \quad \theta = 270^\circ$$

$$x = 270^\circ$$

**Exercise:** Solve each equation for  $0^\circ \leq \theta \leq 360^\circ$ .

(a)  $\operatorname{cosec}^2 \theta = 4$

$$\frac{1}{\sin^2 \theta} = 4$$

$$\sin^2 \theta = \frac{1}{4}$$

$$\sin \theta = \pm \sqrt{\frac{1}{4}}$$

$$\sin \theta = \pm \frac{1}{2}$$

•  $\sin$  is positive in Q1 & Q2

$\sin$  is negative in Q3 & Q4

•  $\theta = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

• in Q1  $\rightarrow \theta = 30^\circ$

in Q2  $\rightarrow \theta = 180^\circ - 30^\circ = 150^\circ$

in Q3  $\rightarrow \theta = 180^\circ + 30^\circ = 210^\circ$

in Q4  $\rightarrow \theta = 360^\circ - 30^\circ = 330^\circ$

$$\therefore \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

(b)  $\tan^2 \theta + 2 \tan \theta + 1 = 0$  let  $u = \tan \theta$

$$u^2 + 2u + 1 = 0$$

$$(u+1)(u+1) = 0$$

$$(u+1)^2 = 0$$

$$u+1 = 0$$

$$u = -1$$

$$\tan \theta = -1$$

•  $\tan$  is negative in Q2 & Q4

•  $\theta = \tan^{-1}(1) = 45^\circ$

• in Q2  $\rightarrow \theta = 180 - 45 = 135^\circ$

in Q4  $\rightarrow \theta = 360 - 45 = 315^\circ$

$$\therefore \theta = 135^\circ, 315^\circ$$